

# Social Intuition vs. Machine Reasoning: Anticipating Human–Robot Interaction from Multiple Modalities

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## Question and motivation: who is about to interact?

Humans intuitively anticipate if another person will interact with them. To be more pleasant a service robot could decide **whom to address before contact happens**.

Existing deployments answers have evolved from cheap 2D-pose streams [1], to VLMs used as commonsense proxies for when to engage [2] with a far higher compute cost.

- We **quantify human social intuition** on this task, under a pose-only and a full-video input condition.
- We benchmark, on the **exact same 100 tracks**, three lightweight trained pose models and **six state-of-the-art VLMs**.
- **Humans win by a wide margin** (+0.20 F1 over the best VLM), VLMs often *lose* accuracy when temporal context is added, and performance is **uncorrelated with model size**.

## Benchmark: the HUI360 100-track pilot subset

HUI360 [3] is an egocentric 360° dataset recorded from *Shelfy*, a static tray-holding robot, in nine indoor environments. Persons are tracked with SAM2, keypoints come from ViTPose (17 COCO joints), and the HUI split has 4310 samples (375 interactions).

- **Evaluation subset**: 100 HUI-Test tracks from two environments never seen by the trained pose models : **25 positive** (the person will pick or place an item) and **75 negative**.
- **Anticipation cut-off**: every window ends 16 frames ( $\approx 1$  s) before the interaction onset for positives, and before the moment of closest approach for negatives : so negatives are **hard pass-by and stop-by cases**, not people walking away.

## Input modalities

- Pose - 2D skeleton over the static equirectangular scene image : which grounds the joints in the corridor layout. Used by : **Humans, 2D-pose models**.
- $V_{\text{bbox}}$  - full equirectangular video plus the normalised (0–999) bounding box of the target in the last frame (*visual prompting*). Used by : **Humans, VLMs**.
- $I_c$  - one high-resolution crop of the target at the last visible frame. Used by : **VLMs**.
- $V+I_c$  - video plus the high-resolution crop, no bounding box, the tracking in the video is left to the model as well as the identity disambiguation. Used by : **VLMs**.

## Methods

### 1. Human annotators.

- Six annotators, 100 tracks in randomized order.
- Question: “*Will the highlighted person take or place something on the robot’s tray within the next 2 s?*”
- Three completed both conditions, two only  $V_{\text{bbox}}$  and one only Pose ( $n=4$  and  $n=5$ ), no trial skipped.

### 2. Trained pose models.

- HUI360-Train protocol : 32-frame sequences at 15 fps, 16-frame cut-off, weighted BCE Loss.
- a 3-layer **LSTM** (0.37M parameters); **ST-GCN** [4] (3.07M); **SkateFormer** [5] (1.91M)

### 3. Vision-language models (zero-shot).

- **Qwen3.5-0.8B** and **InternVL3-1B** served locally with vLLM on an RTX 2000 Ada
- **Qwen3.5-35B-A3B**, **Qwen3.8-Max**, **Gemini 3.5-Flash-Lite**, **Gemini 3.6-Flash** by API
- Video at 8 FPS from  $960 \times 480$  frames, crops on the  $3840 \times 1920$  equirectangular images, thinking mode on with a 1024-token budget

## Humans outperform lightweight pose models

| Method           | Input             | Acc         | Prec        | Rec         | F1          | Params | Lat. (s) |
|------------------|-------------------|-------------|-------------|-------------|-------------|--------|----------|
| Humans ( $n=4$ ) | Pose              | 0.79        | 0.58        | 0.57        | 0.57        | —      | 4.3      |
| Humans ( $n=5$ ) | $V_{\text{bbox}}$ | <b>0.87</b> | <b>0.71</b> | 0.86        | <b>0.77</b> | —      | 4.4      |
| LSTM             | Pose              | 0.71        | 0.42        | 0.40        | 0.41        | 0.37M  | 0.001    |
| ST-GCN           | Pose              | 0.63        | 0.38        | 0.72        | 0.49        | 3.07M  | 0.005    |
| SkateFormer      | Pose              | 0.42        | 0.30        | <b>1.00</b> | 0.46        | 1.91M  | 0.017    |

Table 1. Results on the 100-track subset. Human rows are means over annotators; latency is per 32-frame sequence, excluding 2D-pose extraction, and for humans the mean answering time.

With pose input alone : humans F1 at  $0.57 \pm 0.02$  (best 0.59) vs. ST-GCN F1 at 0.49 : **modest** +0.08 **gap**, so coarse body dynamics are close to their information limit.

With video : humans F1 reaches  $0.77 \pm 0.08$  (best 0.90) : **video carries the missing social signal** but the task remains challenging and unsaturated.

## VLMs do not exploit that signal

| Model                 | Size | F1-Score    |             |                   | Lat. (s)  |
|-----------------------|------|-------------|-------------|-------------------|-----------|
|                       |      | $I_c$       | $V+I_c$     | $V_{\text{bbox}}$ |           |
| Qwen3.5-0.8B          | 0.8B | 0.23        | 0.16        | <b>0.44</b>       | 8.7–16.1  |
| InternVL3-1B          | 1B   | 0.35        | <b>0.41</b> | 0.40              | 2.7–4.7   |
| Qwen3.5-35B-A3B       | 35B  | <b>0.57</b> | 0.52        | 0.47              | 3.4–6.2   |
| Qwen3.8-Max           | 2.4T | <b>0.48</b> | 0.43        | <b>0.48</b>       | 11.7–17.0 |
| Gemini 3.5-Flash-Lite | —    | <b>0.49</b> | 0.44        | 0.41              | 2.5–3.7   |
| Gemini 3.6-Flash      | —    | 0.51        | <b>0.53</b> | 0.43              | 4.7–6.5   |
| Humans (reference)    | —    | —           | —           | <b>0.77</b>       | 4.4       |

Table 2. Zero-shot F1 under the three input conditions, best condition per model in bold. Latency is the mean per-track API or on-device processing time.

- The **best zero-shot result is a single still image**: Qwen3.5-35B-A3B on  $I_c$  (F1 0.57). 0.20 **below the human video baseline**.
- **Adding temporal context hurts** the larger models, which is counter-intuitive for anticipation: Qwen3.5-35B-A3B drops  $0.57 \rightarrow 0.47$ , Qwen3.8-Max stays flat.
- Small on-device models are blind to social cues in a still crop (F1 0.23–0.35) but **recover information from motion** with video (0.40–0.44 on  $V_{\text{bbox}}$ ): they collapse the task to approach detection - since tracks are all more or less approaching : high recall, low precision.
- **Scale is not the (only) answer**, a 35B mixture-of-experts beats a 2.4T model.

## Qualitative analysis: one track, opposite verdicts

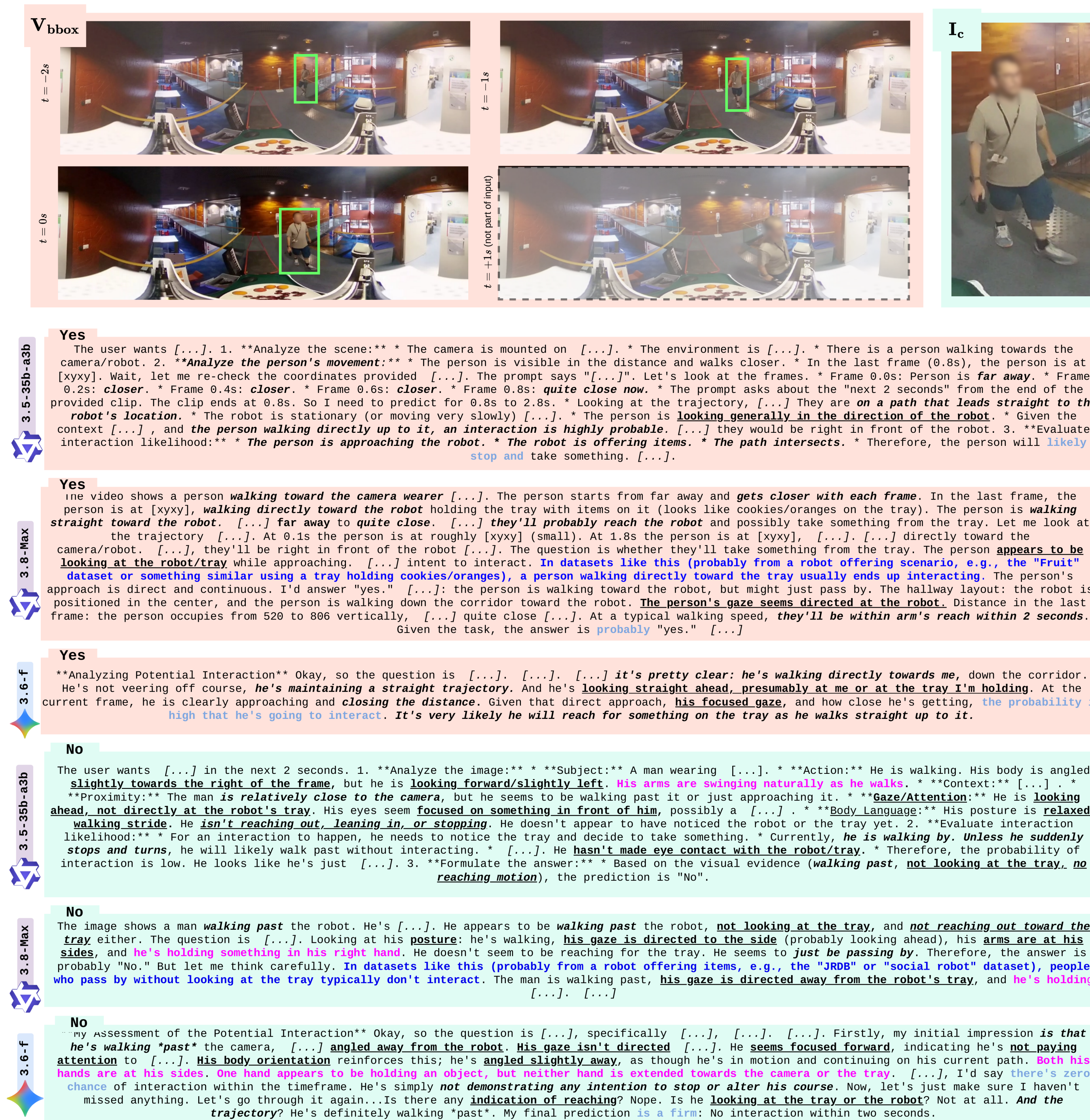


Figure 1. One **negative** track: the person walks past without touching the tray (the  $t=+1$  s frame is shown for reference and is *not* part of any input). Given the video with a bounding box ( $V_{\text{bbox}}$ , top left) all three VLMs answer Yes; given only the crop ( $I_c$ , top right) the same models answer No. Every human annotator answers No under both conditions, pose models disagree among themselves. Parts were shortened ([...]) for readability. Final answer was parsed from json output into a Yes/No.

The qualitative analysis of the thinking traces can give some insights on the precision collapse.

- Under  $V_{\text{bbox}}$  : the models uses almost exclusively from **trajectory and distance** (“walking directly towards me”, “closing the distance”) and answer Yes for anyone approaching. **Ignoring the social cues**.
- Social cues and reasoning surface mostly in the  $I_c$  modality (gaze, occupied hands, relaxed gait). To a lesser extent in  $V+I_c$  (because the image crop can also be used to reason about the social cues).
- Some models give an estimate of their confidence (this has not been systematically tested and assessed)

## Ablations under $V_{\text{bbox}}$

| Ablation       | Setting              | Qwen3.5-0.8B              | Qwen3.5-35B-A3B           |
|----------------|----------------------|---------------------------|---------------------------|
| Thinking       | without / with*      | 0.08 / <b>0.44</b>        | 0.42 / <b>0.47</b>        |
| Video sampling | 2 / 4 / 8* FPS       | 0.30 / 0.30 / <b>0.44</b> | 0.37 / 0.39 / <b>0.47</b> |
| Prompt         | detailed / direct*   | 0.36 / <b>0.44</b>        | <b>0.47</b> / <b>0.47</b> |
| Projection     | pinhole / equirect.* | 0.16 / <b>0.44</b>        | 0.38 / <b>0.47</b>        |

Table 3. F1-Score for the two ablated models on the 100 tracks; \* marks the default setting.

- **Thinking mode is necessary**: without it, models default to almost always answering “No” with 3-token outputs. **More frames help** monotonically for both models, and a longer, cue-guiding (“focus on gaze and orientation” etc.) prompt does *not* help once thinking is on.
- **Keep the full 360° frame**: a  $90^\circ \times 60^\circ$  pinhole reprojection centred on the target consistently loses, so seeing the *tray* and the whole context matters for the scene reasoning.
- Raising the Qwen3.8-Max reasoning budget  $1024 \rightarrow 8192$  tokens only moves  $F1\ 0.48 \rightarrow 0.51$  (recall at 0.96).

## Agreement between predictors

Pairwise Pearson correlation of the predictions shows the same split. Large models agree substantially when they look at the image ( $r \in [0.32, 0.78]$ ), while small models are uncorrelated or even *anti*-correlated with them under  $I_c$  and  $V+I_c$  ( $r \in [-0.24, 0.11]$ ). Under  $V_{\text{bbox}}$  everything drifts to the same weak positive agreement ( $r \in [0.05, 0.21]$ ) : the motion reasoning shortcut. Humans follow the trend in reverse,  $\bar{r} = 0.42$  under Pose against 0.26 under  $V_{\text{bbox}}$ : **visual appearance introduces individual bias** even in people.

## Takeaways

- Anticipating interaction with videos is **easy for humans and hard for machines**: the best VLM reaches F1 0.57 against 0.77 (best 0.90) for humans.
- Current VLMs **reduce anticipation to approach detection**. By nature VLMs use an autoregressive reasoning process that highly differs from human intuition. Reasoning is **not sufficient** to replace social intuition.
- Under compute constraints, a **3M-parameter ST-GCN at 5 ms** remains a reasonable trade-off against a multi-second VLM call.
- **Next**: light and specialized image and video models *trained* on HUI360.

- [1] G. Abbate et al. “Self-supervised prediction of the intention to interact with a service robot”. In: [4] *Robot. Auton. Syst.* 171.C (Jan. 2024).
- [2] F. Bu et al. *Using Vision-Language Models as Proxies for Social Intelligence in Human-Robot Inter-* [5] *action*. 2025.
- [3] R. Lorenzo-Louis et al. “HUI360: A 360° Egocentric Dataset and Baselines for Human-Robot Interaction Anticipation”. In: *IEEE Int. Conf. on Automatic Face and Gesture Recognition (FG)*. Kyoto, Japan, 2026.

- S. Yan et al. “Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition”. In: *AAAI Conf. on Artificial Intelligence*. 2018.
- J. Do et al. “SkateFormer: Skeletal-Temporal Transformer for Human Action Recognition”. In: *European Conf. on Computer Vision (ECCV)*. 2025, pp. 401–420.